

In-service corrosion and grain boundary oxidation in neutron-irradiated 316 stainless steel baffle-former bolts

Timothy G. Lach^{1*}, Maxim N. Gussev², and Xiang (Frank) Chen¹

¹Materials Science and Technology Division, Oak Ridge National Laboratory, 1 Bethel Valley Road, Oak Ridge, TN 37831, USA

²Nuclear Energy and Fuel Cycle Division, Oak Ridge National Laboratory, 1 Bethel Valley Road, Oak Ridge, TN 37831, USA

* Corresponding author: lachtg@ornl.gov

ABSTRACT

Reactor core internal components such as baffle-former bolts (BFBs) are subjected to significant mechanical stress, corrosive environment, and neutron irradiation from the reactor core during the plant operation. Over the long operation period, these conditions lead to potential degradation and of the bolts. In this work, characterization was performed on the oxidized surface of stainless steel BFBs harvested from a commercial pressurized water reactor (PWR) after 40 years of operation. The analysis shows that a complex multilayered surface oxide with six identified layers formed that is different from 2-layer structure commonly observed in model experiments. The oxide varies by composition – predominantly Fe, Cr, and Ni, grain size, and phase, and has features resembling both unirradiated and radiation/ corrosion experiments likely due to the low radiation flux compared to ion-irradiation or the test reactor radiation. In addition, grain boundary oxidative attack featured a pathway for Fe and other elements to move from the metal matrix to the outermost oxide. The results help assess PWR lifetime extension, put into context previous experimental studies, and provide input for designing experiments combining radiation and corrosion effects.

Notice: This manuscript has been authored by UT-Battelle, LLC under Contract No. DE-AC05-00OR22725 with the U.S. Department of Energy. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this manuscript, or allow others to do so, for United States Government purposes. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (<http://energy.gov/downloads/doe-public-access-plan>).

Keywords: Baffle-former bolt; In-service irradiation and corrosion; Multilayered oxide structure; Radiation-induced segregation (RIS); Irradiation assisted stress corrosion cracking (IASCC); Crack initiation

1. Introduction

Light water reactor (LWR) internal components are exposed to multiple extreme environmental conditions such as high temperature water, neutron and gamma radiation, and mechanical stresses. These conditions leave the materials susceptible to microstructural changes that increase the susceptibility to void swelling, embrittlement, and irradiation-assisted stress corrosion cracking (IASCC). In particular, IASCC is of high concern for life extension of pressurized water reactors

(PWRs) [1-3]. To understand IASCC, the effects of irradiation on both general corrosion and localized grain boundary oxidation must be studied.

Most experimental research into IASCC and irradiation/corrosion has focused on sequential exposure of irradiated steel to corrosive environments such as simulated PWR water [4-10] or relatively short-term combined ion (usually proton) irradiation-corrosion experiments [11-13]. Being valuable from scientific and engineering points of view, out-of-reactor short-term experiments cannot fully reproduce in-service conditions. Duration of the corrosion processes is an important parameter; irradiation accelerates corrosion at relatively short time scales, but potentially reduces corrosion rates at longer time scales due to the evolution of the Cr content in the inner oxide layer [7, 8]. Also, model experiments often do not include the effect of water radiolysis observed during concurrent irradiation/corrosion experiments [11-14]. Radiolysis has been found to increase the corrosion potential of the water, thus dissolving the inner Cr-rich oxide and reducing its passivating ability but still reducing the overall corrosion rate compared to sequential exposure [8]. Additionally, parameters like flux (damage rate), neutron spectra, exposure time are difficult to reproduce. It underlines the importance of studying harvested in-service components, which have experienced LWR conditions with concurrent irradiation, corrosion, and mechanical stresses.

As one of the PWR internal components, baffle-former bolts (BFBs) are particularly susceptible to IASCC attack. Over the long operation period, the extreme conditions in operating reactors can lead to effects like embrittlement, creep, stress relaxation followed by occasional overstressing, and large-scale cracking. The first degraded BFBs were observed in 1989 [15]. In support of evaluating long-term operation performance of materials used in core internal components, the Materials Research Pathway under the U.S. Department of Energy (DOE), Office of Nuclear Energy, Light Water Reactor Sustainability Program pursued the retrieval of aged structural components such as BFBs for the study of the microstructure, mechanical, and corrosion-related properties. Two high fluence BFBs were removed in 2011 from a Westinghouse two-loop downflow type PWR and delivered to Oak Ridge National Laboratory. The BFBs showed no indication of cracking by visual and ultrasonic inspection after being withdrawn from service. In addition to standard post irradiation examination (PIE) to analyze radiation-induced microstructural and mechanical property evolution along the length of the bolts found in previous work [16] and ongoing work, the oxidized surfaces of the BFBs are also the subject of high interest.

This study focuses on the curved region of bolt head/shank intersection. This intersection still has the in-service exposure surface intact from handling, which is of high interest because it has a high mechanical stress concentration compared to the rest of the bolt head or shank and was also exposed to the primary water [3]. Thus, this region has the three primary components for IASCC: corrosive environment, stress, and irradiation. To the authors' knowledge, this is the first time the native oxide of in-service BFBs has been characterized with high resolution analytical electron microscopy. Characterization of the oxidized surface and oxidized grain boundary will be discussed in detail. The results show that while simulated conditions in sequential experiments or concurrent ion irradiation/corrosion experiments do provide useful information about the corrosion and grain boundary oxidation behavior in stainless steels, the oxidation behavior observed in materials after PWR in-service exposure is more complex.

2. Materials and Methods

Two BFBs studied in the present work were removed from a Westinghouse two-loop downflow type pressurized water reactor (PWR) in 2011. During routine inspection, the BFBs showed no visual indication of cracking, though a lower torque was needed for removal from the baffle structure than the original torque specified during installation, suggesting radiation-induced stress relaxation [17]. Importantly, the bolts do still have some of the original surface that was exposed to coolant water where the bolt head intersects with the bolt shank. Optical and microscopy survey showed that this area did not receive significant mechanical damage during removal and handling being protected, at some degree, by the bolt geometry.

The element composition of bolts are within the specifications for type 316-SS based on atom probe tomography analysis conducted previously [16]. The bolts were in the reactor for approximately 40 years; the calculated radiation dose of each bolt was different as the bolt head in #4412 received ~41 displacements per atom (dpa) while #4416 received ~29 dpa due to the difference in neutron flux at each bolt location. The exact temperature profiles of the bolts are unknown, but it is known that the power plant went through 35 operating cycles for fuel replacement and maintenance [18], so the temperature did ramp up and down. The operating temperature at this portion of the bolts was likely about 305-317 °C based on finite element modeling [3]. After removal from service, the bolts were cut inside a hot cell, and the bolt heads were sliced further to produce samples suitable for out-of-hot-cell handling, Figure 1.

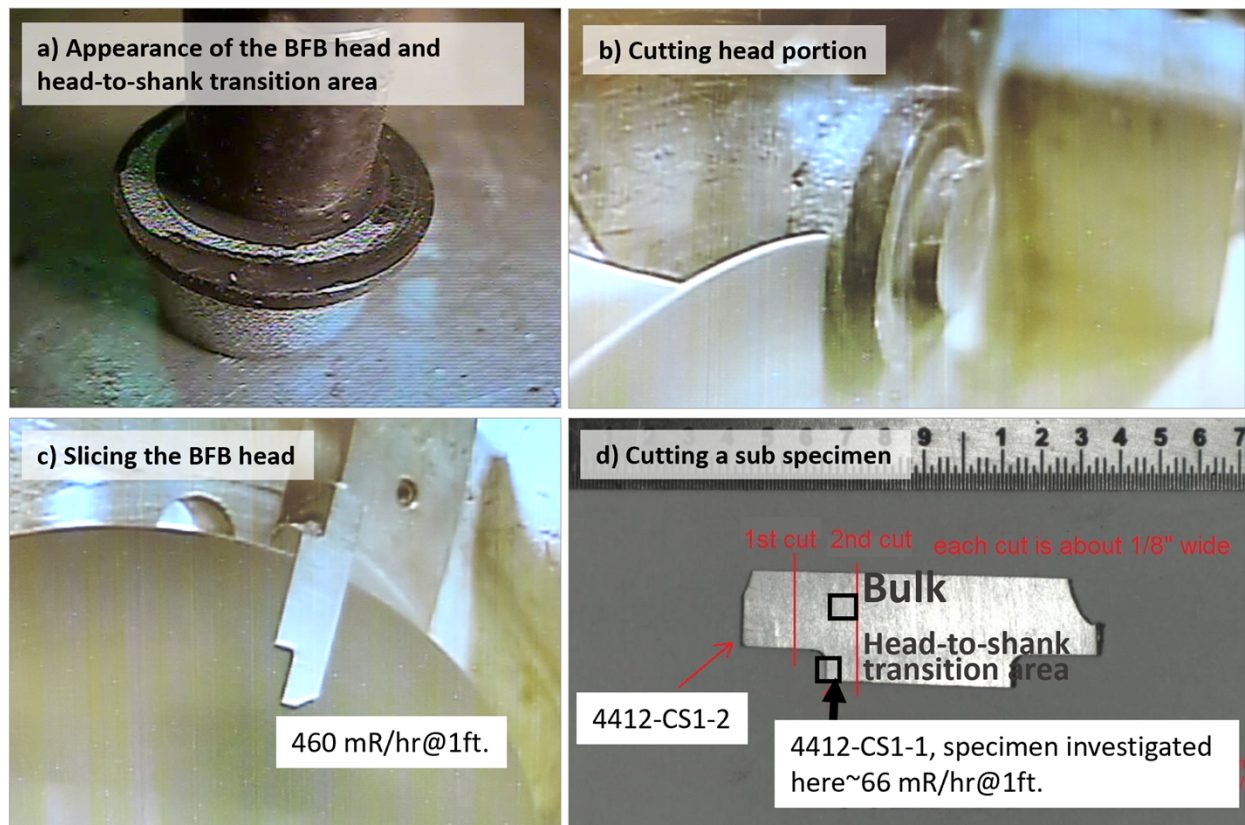


Figure 1. a) Harvested baffle bolt; b, c) Cutting and slicing the BFB head; d) Location of the studied specimen.

A slice of the bolt head sections for both bolts was polished to a mirror-finish with a final polishing step using colloidal silica in a vibratory polisher. The cross-section of the near-surface area in the bolt head/shank intersection curved region was characterized by SEM-electron backscatter diffraction (EBSD) using a Tescan Mira3 SEM equipped with an Oxford Instruments Symmetry S2 high speed EBSD detector. SEM voltage was 20 kV; scanning was performed on a rectangular grid with step size 500 nm (for general view scan) and 125 nm (for high-resolution scans).

Transmission electron microscopy (TEM) liftouts normal to the surface for further analysis were carried out using a Thermo Fisher Versa3D dual-beam focused ion-beam (FIB)/SEM. The TEM sample preparation followed the procedures of initial trenching, cutting, and thinning to 200 nm with 30 keV Ga ion beam, rough polishing to 100 nm with ion beam energy gradually reduced from 16 keV to 8 and 5 keV, and fine polishing with 2 keV ion beam to remove most FIB-induced damage. Before loading into the TEM, samples were always cleaned with a Fischione 1020 Nanomill for 10 minutes on each side using 900 eV Ar⁺ ions. Microstructural and elemental mapping characterization was performed using transmission Kikuchi diffraction (tKD) using the Tescan Mira3 SEM and scanning TEM (STEM)/Energy-dispersive X-ray spectroscopy (EDS)-based mapping using a Thermo Fisher Talos F200X STEM equipped with the SuperX 4-sector X-ray EDS system. This was done to identify and quantify structural defects, elemental segregation, and nanoscale precipitates. The SEM-EDS, EBSD, and tKD analysis and mapping were performed using the Oxford Instruments Aztec Crystal and EDAX OIM v.8 software packages. The TEM and STEM-EDS analysis was performed using Thermo Fisher Velox software.

3. Results

3.1. General Baffle Former Bolt microstructure

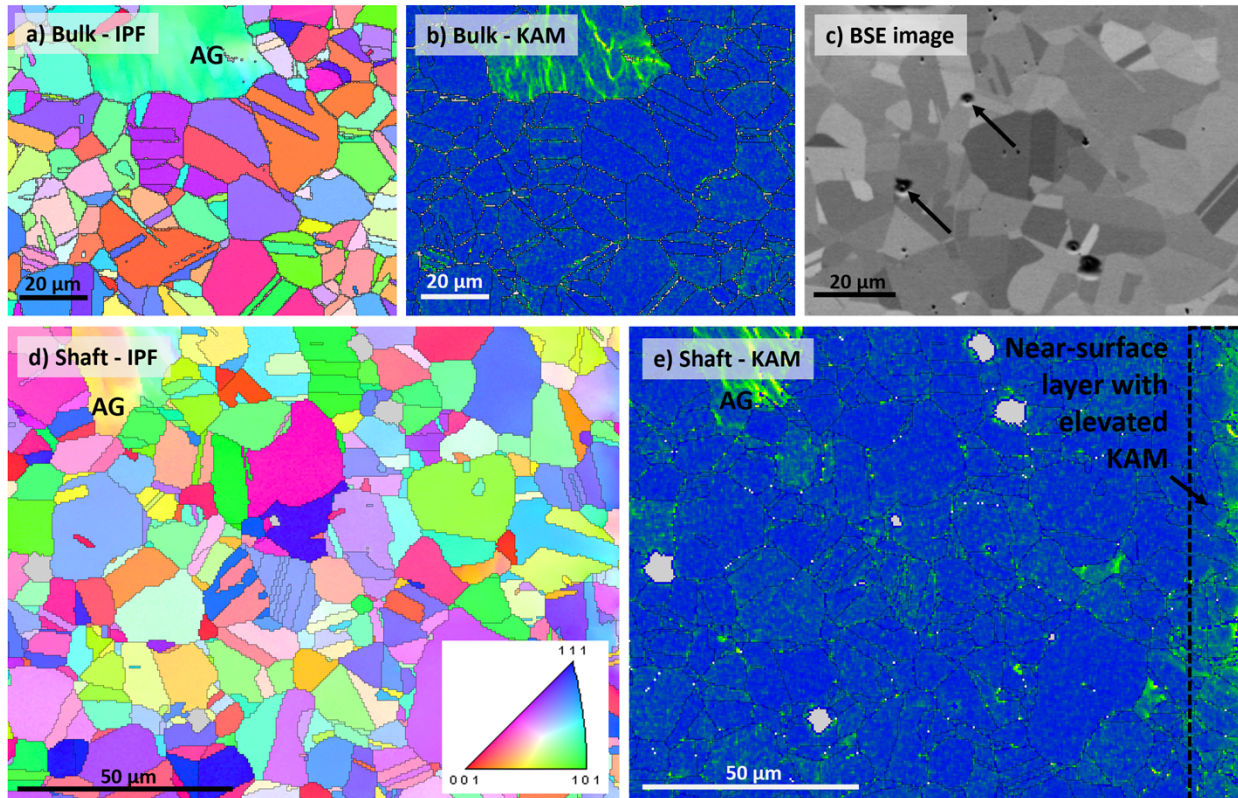


Figure 2. General microstructure of the BFB #4412 head portion. a, b) EBSD inverse pole figure (IPF) and KAM maps, respectively taken at the distance of approximately 2 mm from the head-to-shaft-transition (the area is marked as “Bulk” in Figure 1); c) SEM BSE image taken close to the shaft and illustrating grain morphology and presence of non-metallic inclusions (black arrows). d, e) EBSD maps recorded close to the shaft ((the area is marked as “Head-to-shank transition” in Figure 1). The IPFs are colored in the horizontal direction. EBSD step size: 500 nm.

Figure 2(a-c) shows a representative BFB head microstructure recorded in the head area (approximately 2 mm from the head-to-shaft transition). Equiaxial well-shaped grains dominate the microstructure; many grains reveal annealing twins. No retained ferrite was observed in the microstructure, so no phase map is shown. The SEM BSE image, Figure 2(c), shows the presence of nonmetallic inclusions of a few micrometers in size.

Although EBSD data suggest a mostly annealed microstructure with minor misorientation gradients, some grains contain slightly elevated misorientation levels with multiple in-grain low-angle boundaries, see the kernel average misorientation (KAM) map, Figure 2(a, b). Several of such grains are marked as “AG” (“abnormal grains”). A population of much smaller recrystallized grains fully surrounds AGs, ruling out the possibility that cold work (occasional or intentional deformation) was a possible source of the elevated in-grain misorientation levels. KAM distribution inside such abnormal grains is also not typical for cold or warm deformation in reference (non-irradiated) steel or strain localization processes in irradiated steel. The specific shape of grain boundaries (arcs or semicircles) for AGs suggests that these grains may be a result of abnormal grain growth [19] (sometimes termed also as “anomalous grain growth”).

Figure 2(d, e) shows the microstructure of the head-to-shank transition area. One may see the same pattern of dominating well-shaped equiaxed grains mixed with relatively rare AGs. Also, there is a thin layer (shown by dotted rectangle in Figure 2(e) along the edge with a thickness estimated at 10–15 μm with elevated KAM values. Elevated local misorientation means high local dislocation density along the edge. The impact on general corrosion of a deformed surface layer could be an enhancement of uniform oxidation [20–22]; however, this prior work showed an ultrafine grain (UFG) microstructure at the surface, which appears to be absent here, although it cannot be entirely ruled out since the initial surface condition of the bolts is unknown. As will be shown in Section 3.3, the nanograin oxide layer, if from a consumed UFG surface, is much thinner than what would be found from machining [22]. The impact on IASCC may be significant due to the importance of strain localization on IASCC [1] and will be discussed in more details in Sections 3.4 and 4.2. Prior research has shown that a highly deformed and/or UFG surface may have enhanced SCC resistance in unirradiated environments [20, 21]. This thin layer is likely related to BFB manufacturing.

Not shown here but presented elsewhere [16, 23] is the radiation-induced defect structures. There is a high density of nanocavities with limited swelling, dislocation loops, and Ni/Si-rich precipitates within the grains and radiation-induced segregation (RIS) to grain boundaries. These structures have a substantive impact on radiation hardening and IASCC susceptibility. A detailed analysis of as-irradiated microstructure and post-service mechanical performance will be given in a separate focused paper.

3.2. Appearance of the water-exposed surface

As was mentioned above, the slices still had the water exposed surface retained, Figure 3. The oxidized surface is heterogeneous in nature with large lighter shade features down to smaller nodules. The range in size of the nodule-like features is large with some greater than 10 μm across

and others less than 1 μm . The surface microstructure may be interpreted as an oxide film with coarse particles of iron-rich oxide. Such a structure is common and was often observed in corrosion and stress corrosion cracking experiments [24-26].

In this region of the bolts (head-to-shank transition area), there are also cracks in the oxide layer, visible in SEM, while there are no cracks visible away from the curved region of the bolt. Some of the cracks appear to be over 20 μm in length. Most cracks are oriented in the BFB axial direction, likely reflecting the direction of acting stress.

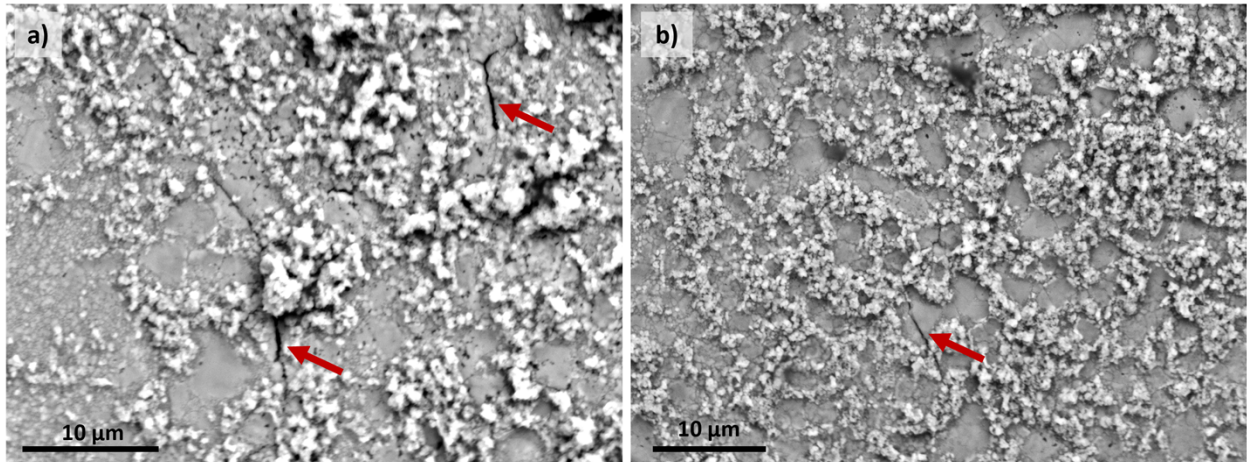
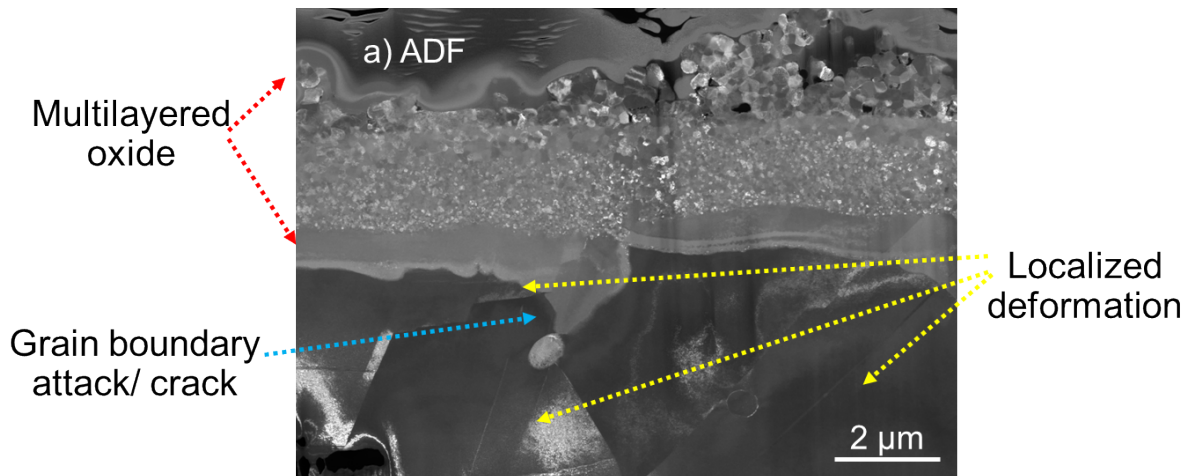


Figure 3: Backscatter electron SEM images from two different areas of the surface of the head/shank transition area of BFB #4412. There are multiple small cracks visible (red arrows) and rough heterogeneously sized Fe-rich oxide particles across surface.

3.3. Complexity of the surface oxide multilayer structure

To investigate further the oxidation behavior and grain boundary attack at the cracks, targeted FIB liftouts were taken from the oxidized surface targeting cracks. One such liftout is shown in Figure 4 for BFB #4412, highlighting specific features: complex surface oxidation, grain boundary oxidative attack and cracking, and localized deformation in the form of dislocation channels [23].



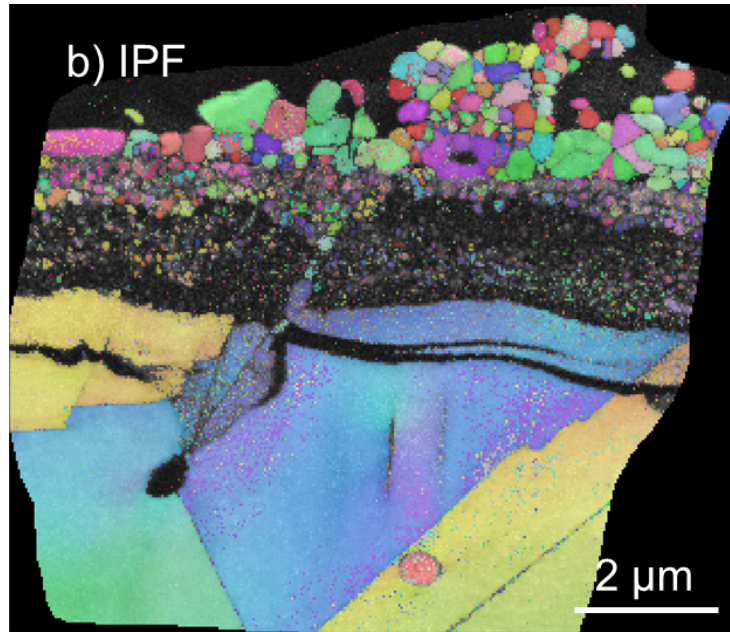


Figure 4: a) STEM-ADF image of liftout from BFB #4412 taken from oxidized surface near “crack #4” in Figure 9. b) tKD IPF map of the same liftout. Multiple features are apparent including a multilayered oxide (red arrows), grain boundary attack (blue arrow), and localized deformation features inside the bolt (yellow arrows). The IPF map is based on the face centered cubic (FCC) lattice showing the matrix austenite and oxide layers have same lattice.

The surface oxide on the curved region of the BFBs is complex. The STEM images, tKD IPF map, and EDS map for BFB #4412 in Figure 5, the TEM and selected area electron diffraction (SAED) patterns for BFB #4412 in Figure 6, STEM-EDS analysis for BFB #4412 in Figure 7, and STEM-EDS analysis for BFB #4416 in Figure 8 highlight six distinct layers of the surface oxide that separate themselves by composition and by grain size. The observed porosity within the oxide layers of the liftouts is minimal. Interestingly, instead of six layers, only three layers are commonly identified for corrosion and radiation-enhanced corrosion [7, 13]. The total oxide thickness for this BFB is $\sim 3 \mu\text{m}$ near its thinnest point and up to $\sim 5 \mu\text{m}$ near grain boundaries.

The outermost oxide layer, denoted as layer 1, is an Fe-rich oxide with large grains $\sim 250\text{-}500 \text{ nm}$ in size. The thickness of this layer varies from one grain thick ($\sim 250 \text{ nm}$) to multiple grains but seems to be largest around grain boundaries. The second layer is an intermixed Fe/Ni-rich and Cr-rich oxide with grain sizes of $\sim 100\text{-}200 \text{ nm}$ and a thickness of approximately $600\text{-}700 \text{ nm}$. The grains appear to be Fe/Ni-rich while the oxide boundaries are Cr-rich. The third layer is similar in composition to layer 2 but is marked by a much finer grain size of well under 100 nm ($\sim 50 \text{ nm}$). The interesting segregation behavior of what appears as Fe/Ni-rich grain centers surrounded by Cr is more obvious in layer 2 but also appears to be the case in the smaller grains in layer 3. The variation in composition from oxide grain to oxide grain can clearly be seen in Figure 7(f). For this BFB #4412, layer 3 is $\sim 1.3 \mu\text{m}$ thick, but as is shown in Figure 8, the layer 3 oxide thickness can vary substantially from bolt to bolt with only $\sim 150\text{-}400 \text{ nm}$ for the sample taken from BFB #4416. While layer 1 may traditionally be referred to as the outer oxide, layers 1, 2, and 3 here are all outside the original metal surface and so could be considered outer oxide layers. This is based on the grain size being much less than the inner oxide layers 4 and 5 as well as a growing mismatch in the zone axis orientation of the base metal and inner oxide layers 4 and 5 in Figure 6.

The fourth and fifth oxide layers appear monolithic in nature but are segregated by composition with the fourth layer being Cr/Fe/Ni-rich while the fifth layer closest to the alloy surface is depleted in Fe and Ni. The “monolithic” nature is a result of these layers being originally the base metal substrate and thus part of the original large micron-scale grains. This is clear based on two observations: the tKD-IPF map in Figure 4 shows that layer 4 has the same orientation as the base metal below (layer 5 could not be indexed) just as was observed previously in high temperature oxidation of a Fe-19Cr-9Ni alloy [27], and the TEM BF image and selected area electron diffraction (SAED) patterns in Figure 6 showing the layer 4 contrast matches that of the base metal below when tilted to a [100] zone axis and the diffraction pattern matching orientation between the metal and oxide layers of 4 and 5, though all three – base metal and oxide layers 4/5 – have different crystal structures. Looking at the SAED patterns, layer 5 looks to have two sets of diffraction spots – ones that match the base metal (possibly due to an overlap of the selected area aperture with the base metal) and ones that have a similar but slightly smaller spacing with diffraction spots inside the base metal diffraction spots, consistent with a rock salt structure with a cube-on-cube orientation relationship with the base metal [7]; layer 4 has the same single crystalline pattern but with added superlattice reflections consistent with a spinel structure. The original metal surface is at the boundary between the nanocrystalline layer 3 and layer 4.

The final layer (layer 6) is a thin sub-5 nm Ni-enriched layer at the matrix/oxide interface. This Ni-enriched layer has been observed in experimental conditions in both sequential and concurrent testing [6, 28]. However, interestingly, this Ni-layer is not observed on the right side of the grain boundary attack, as shown in the linear concentration profiles in Figure 7. This may be due to the behavior of the grain boundary itself, as it breaks up the layers nearest the matrix or the Ni segregation to the grain boundary discourages the formation of the thin Ni-enriched layer. This layer could not be identified in the tKD maps in Figure 5 or the TEM imaging and SAED diffraction in Figure 6, meaning it is purely an elemental segregation of Ni to the metal/oxide interface.

The composition gradient of each layer, 1 through 5, are shown in 1-D composition profiles in Figure 7(c, d). The thickness of layers 4 and 5 are ~300 nm, which are about the same as in BFB #4416 in Figure 8. The total thickness of the surface oxide in BFB #4416 in Figure 8 is ~1.7-2 μm , or about two-thirds of the oxide thickness for BFB #4412. This could be due to multiple reasons, such as: (1) the thickness of the oxide likely can vary between bolts; (2) the radiation dose is different and could affect oxide formation; and (3) the PWR water conditions and flow could be slightly different.

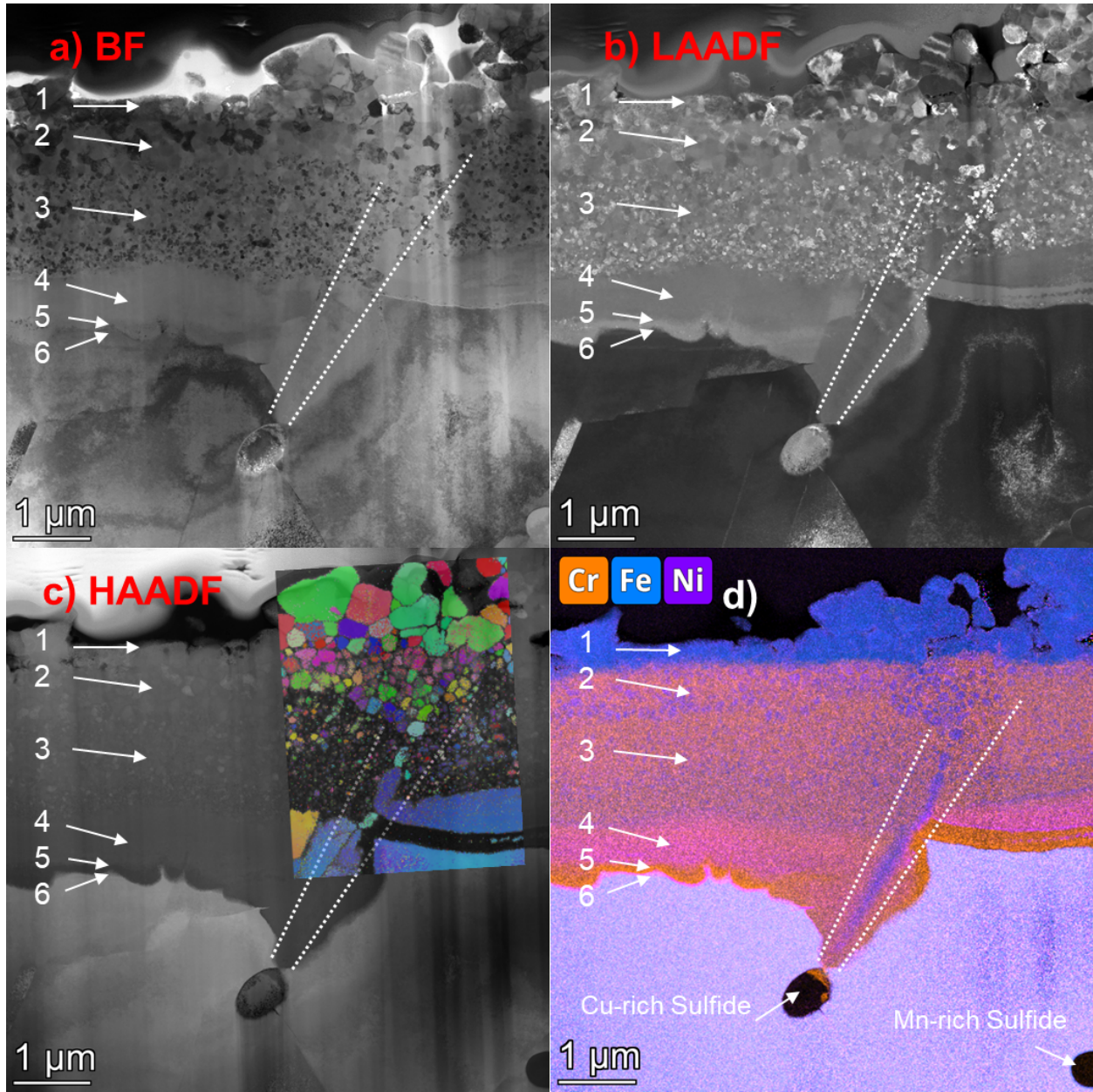


Figure 5: Complexity of the BFB in-service oxidation processes. a) STEM-BF, b) STEM-LAADF, c) STEM-HAADF with tKD-IPF map overlay, and d) STEM-EDS Cr/Fe/Ni map of BFB #4412 highlighting the multilayered oxide and grain boundary attack. Numbers correspond with each oxide layer based on grain size and composition. The two near-parallel dotted lines highlight the grain boundary pathway for the outward diffusion of Fe/Ni. Note: the fit of the tKD-IPF map overlayed in (c) is not perfect as there may have been some drift during the tKD scan compared to the STEM imaging.

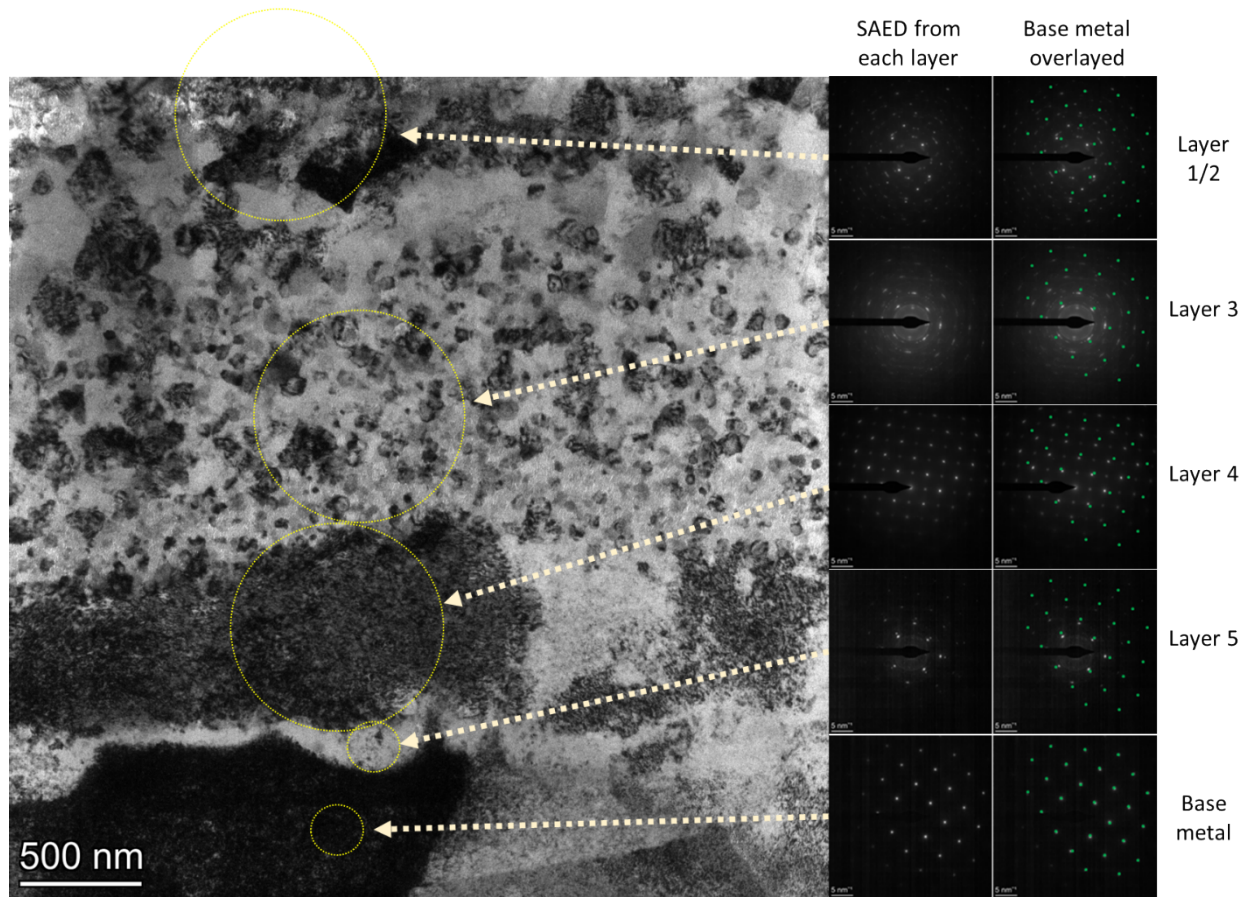
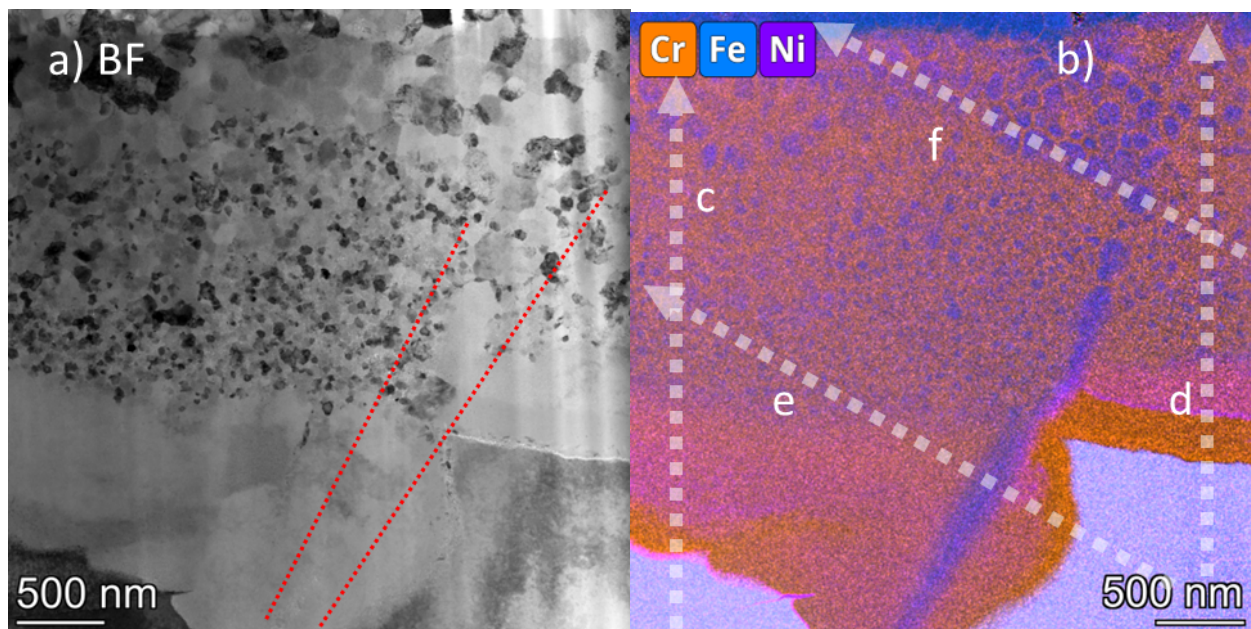


Figure 6: TEM BF image and SAED patterns of surface oxide tilted to $[100]$ zone axis of a grain in the base metal just below the oxide in BFB #4412. Contrast in the BF image and the SAED patterns from layers 4 and 5 highlight that these layers were a part of the original metal, and the original metal surface is between layers 3 and 4.



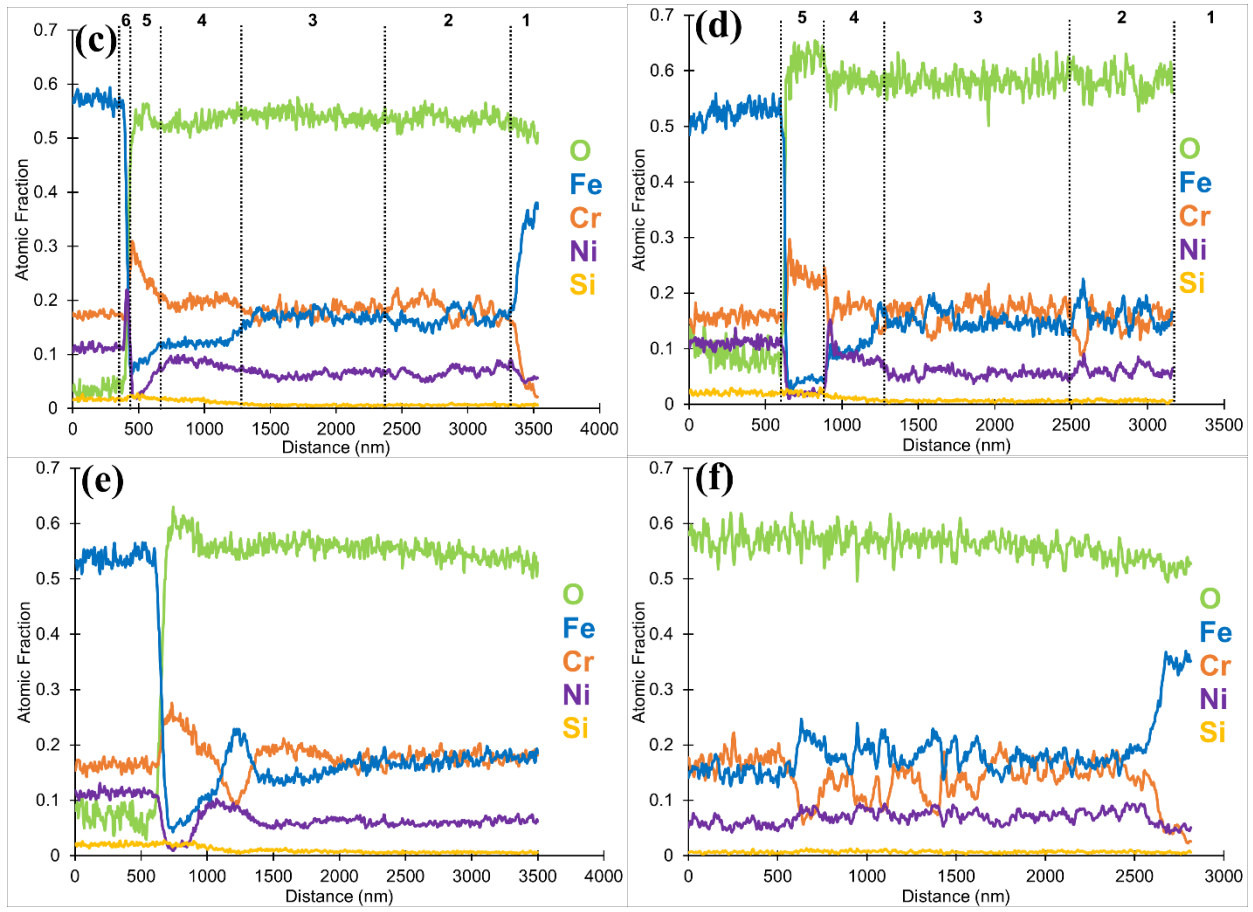
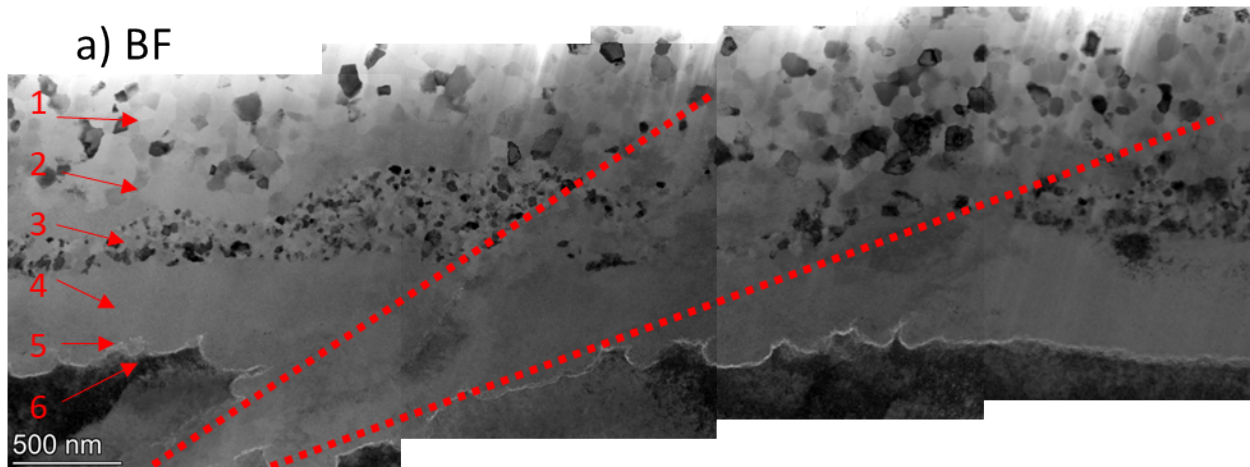


Figure 7: Element distribution profiles through the oxide layers in BFB #4412. a) STEM-BF image and b) STEM-EDS map of Cr/Fe/Ni overlaid showing oxide and grain boundary attack. (c-f) Concentration profiles corresponding to arrows in (b). Profiles normal to the surface (c and d) delineate the boundaries of each layer.



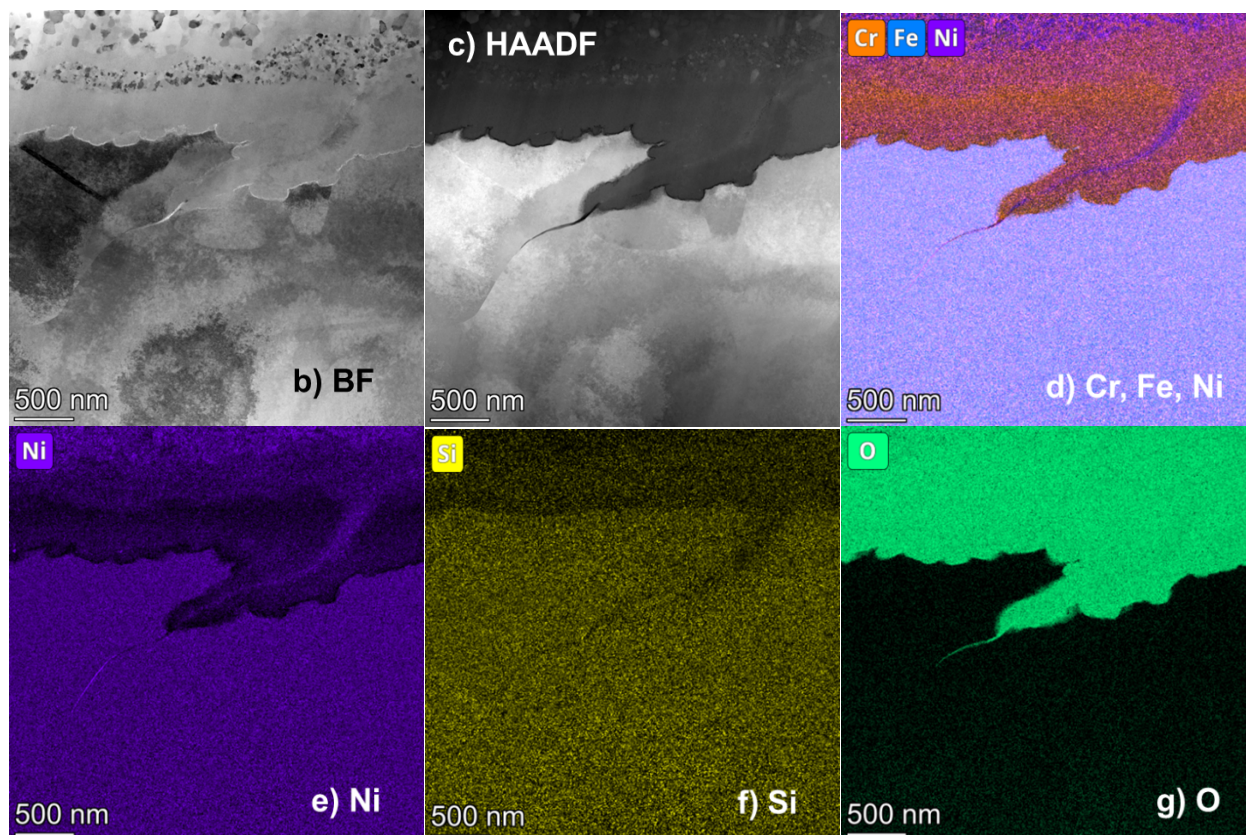


Figure 8: Grain boundary attack and crack initiation in lower dose BFB #4416. a) STEM-BF images near an oxidized grain boundary in BFB #4416 highlighting the multilayered oxide and grain boundary attack. Numbers correspond with each oxide layer based on grain size and composition. b) STEM-BF and c) STEM-HAADF images showing oxide and grain boundary attack. STEM-EDS maps of d) Cr, Fe, Ni overlapped, e) Ni, f) Si, and g) O showing segregation of Ni at grain boundary but a depletion of Si near the oxidized grain boundary.

3.4. Specific microstructure features near the surface and in the oxide layer

3.4.1. Grain boundary corrosion.

Figure 5 and Figure 7 highlight the interesting oxidative grain boundary attack. The near-parallel dotted lines in Figure 5 highlight two features: the grain boundary oxide is enriched in Fe and partially Ni or conversely depleted in Cr, and the oxide grain size is much larger than the oxides in layers 3, 4, and 5 away from the grain boundary. The grain boundary acts as a fast diffusion pathway or pipeline for Fe and Ni to move to the outer oxides, especially layer 1 which is thickest around the grain boundary. The larger grains in layers 1 and 2 along the ‘pipeline’ are all enriched in Fe and Ni as highlighted in the concentration profiles in Figure 7(e, f). This is likely exacerbated by radiation-induced segregation of Ni and Si to the grain boundary and depletion of Cr at the grain boundary. This same behavior observed in BFB #4412 can also be seen in BFB #4416 as layer 3 is clearly interrupted by larger grains near the grain boundary in Figure 8 and the original grain boundary location within the oxide is enriched in Fe and Ni and depleted in Cr. Radiation-induced segregation not only enhances grain boundary attack, but it also enhances external oxide growth. Outside of this ‘pipeline’, oxidation forms a wide wedge shape into the grain boundary. The wedge-shape may form as a means to further propagate the grain boundary oxidation, or possibly due to the grain boundary oxidation being hindered, such as by a sulfide particle like the Cu-rich sulfide particle in Figure 5.

3.4.2. Initial stages of stress corrosion cracking.

SEM imaging, as shown in BFB #4412 in Figure 9(a), demonstrated multiple small cracks along the specimen edge; all observed cracks were located within the curved area, where stress concentration would happen. The longest crack reached $\sim 6 \mu\text{m}$. All analyzed cracks appeared to be filled with oxides. Some but not all cracks originated from oxide pockets where there was wide wedge-shaped oxidation visible. SEM/EBSD crystallography analysis results for one of the cracks are shown in Figure 9(b, c). One may see the crack propagated along random high-angle grain boundary (RHAB). Special $\Sigma 3$ boundaries were immune to crack formation in all analyzed cases, which would correlate with the lack of Ni/Si RIS and Cr depletion at coherent $\Sigma 3$ grain and twin boundaries [29].

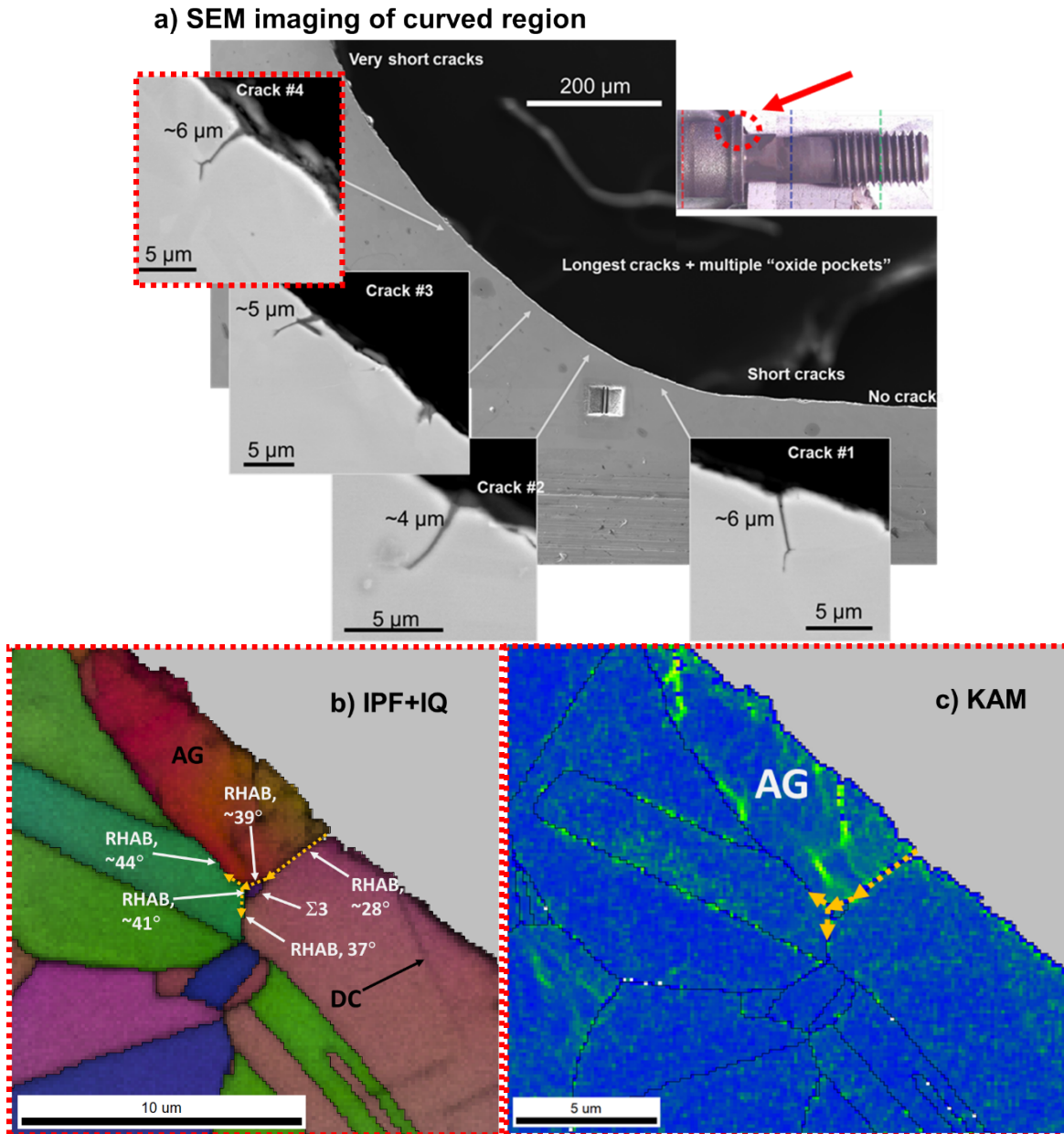


Figure 9: a) SEM images from the curved section of BFB #4412 where the bolt head meets the bolt shank. Microscale cracks indicative of IASCC are found in the curved region and not in the

straight regions. b, c) IPF and KAM analysis of in-service formed short crack highlighted by red dotted box in a). AG: abnormal grain; DC: defect-free channel. EBSD step size: 125 nm.

One specific aspect is important to mention related to IASCC. Multiple defect-free channels were observed in the near-surface grains – see Figure 9(c) and Figure 5(a) – suggesting at least one significant loading event while in service. Channel formation was observed at stress levels likely ranging from 45% to 60–80% of the irradiated yield stress with high stress experienced locally at the channel [1, 30, 31]. The channel density, characterized by the fraction of grains containing defect-free channels and the number of channels per grain, strongly depends on the strain rate or loading rate, temperature, alloy composition, and microstructure. Currently, it is not possible to precisely determine the stress level experienced by the examined baffle former bolt while in service. However, the presence of channels in the majority of analyzed near-surface grains implies that the stress levels approached values comparable to the yield stress of the irradiated material.

Formation of defect-free channels in the investigated baffle former bolts and their complex microstructure are addressed in our separate, focused paper [23]. In the context of the present work, loading event may explain appearance of cracks in the oxide film, Figure 3, and, if stress acted for reasonably long time, formation of short stress corrosion cracks, Figure 9(a). These cracks were not observed in regions of the bolt surface away from the curved region where stress would be highest.

4. Discussion

4.1. General oxidation in commercial PWR environment – comparison with experiments

Surface corrosion of stainless steels in PWR water conditions at around 300 °C, whether with or without radiation, typically is presented as two oxide layers with a Cr-rich inner oxide and an Fe-rich outer oxide [4, 7, 32, 33]. The fully dense, multilayered oxide found at the surface of the harvested BFBs in this study have a similar generalized inner and outer oxides; however, both the inner and outer oxides can be divided into multiple layers themselves. The inner oxide, noted as layers 4 and 5 in Figure 5, are split between a Cr-(Fe,Ni)-oxide with a spinel structure and a Cr-oxide with a rock salt structure nearest the current metal surface. The Cr-(Fe, Ni)-oxide layer 4 seems to function as a compositional intermediary between the outer oxide layers, which have a higher concentration of Fe, and the innermost oxide layer 5. As illustrated in Figure 7, layer 5 consists primarily of Cr oxide. In contrast, layers 2 and 3 contain roughly equal amounts of Cr and Fe, while layer 4 contains more Fe than layer 5, but less than what is found in layers 2 and 3. Rather than just a composition gradient, it is clear the inner oxide is phase separated from the outer oxide.

The outer oxide, on the other hand, has three layers that all have a polycrystalline spinel structure but vary by composition and grain size. Layers 2 and 3 have similar composition and composition distribution in and around the grains but vary strongly by grain size. Layer 3, which formed on the original metal surface, was dominated by oxide grain nucleation rather than grain growth, while layer 2 with the larger grains was dominated by grain growth. It is clearer in the larger grains in layer 2, but in both layers, the grain interiors are Fe and Ni-rich while the grain boundaries are Cr-rich. This elemental separation may limit grain growth in layers 2 and especially 3, and it could be a result of radiation-enhanced diffusion within the oxide possibly due to a vacancy-segregant interaction [34, 35]. The most outward layer, layer 1, is relatively thin throughout – one grain thick – but also is comprised of the large nodules shown in Figure 3. As will be discussed in the next

section, these nodules likely form more readily and are larger over grain boundaries. By analogy with ref. [36], the layers may have different strength, Young modulus, and hardness, but this aspect is out of the present work scope.

The initial oxidation likely began at the interface between layers 3 and 4. Cr, followed by Fe and Ni, move to the metal surface and then moves outward forming an oxide as O moves inward into the metal. Outward Cr diffusion may have been enhanced by both irradiation and by the presence of network dislocations formed during manufacturing [22]. This likely appears to take place rapidly initially based on the nanocrystalline nature of layer 3. However, perhaps as a result of the Cr segregation to the oxide grain boundaries (Figure 7), there may be limited continued grain growth, with more grain growth on the outer layers 1 and 2 where the Cr content is less. Layer 1 in particular is where the Cr reached its limit. On the inner oxide, layer 5 perhaps formed separate from layer 4 as Fe and Ni moved to the outer oxides, depleting layer 5 of Fe and Ni leaving behind a Cr-enriched and protective inner oxide layer, perhaps also enhanced by irradiation [8].

Previous studies have shown that the effects of irradiation on corrosion of austenitic stainless steels depend on whether irradiation and corrosion processes occur concurrently [12] or sequentially [4], and whether the irradiation induces radiolysis within the water [33]. In sequential irradiation followed by corrosion experiments, the degree of corrosion was suppressed in the irradiated material due to radiation enhanced diffusion (RED) of Cr to the surface and forming a more protective inner oxide [7]. The irradiated material also has a thin rock salt structure form at the metal/oxide interface.

In concurrent radiation/corrosion with ion irradiation in PWR water, radiolysis increases the activity of oxygen which enhances oxide formation while destabilizing the inner oxide porosity [33]; with the lack of porosity, the oxidation rate is reduced as it correlates with porosity or excess volume [32]. However, the inner oxide film dissolves readily when in contact with water under radiolysis; so the corrosion rate reaches a steady state when the oxide growth rate equals the dissolution rate on rate [33]. Interestingly, they also found that radiation damage within the steel (not in the oxide) had minimal impact on corrosion in PWR water. However, proton damage rates and radiolysis rates are on the order of 100-150 times higher than in a LWR environment [33].

Comparatively, the morphology, phases, and compositional distribution of the oxide layers in the harvested BFBs appear to borrow features of both irradiated and unirradiated materials from these prior studies into irradiation-affected corrosion. Layer 5, noted from Figure 5 and Figure 6, has the rock salt structure, which was observed in proton irradiated material that was subsequently exposed to PWR-type water [7]. However, the overall oxide thickness and the morphology of the interface between the inner oxide and the metal resemble that of the unirradiated and corroded material [7], as unirradiated material had a thicker oxide and a curvy/ non-flat interface with the metal, and the inner oxide (layers 4-6) match the orientation of the underlying metal grains as shown in Figure 4 similarly to that of oxidation of polished surfaces [27]. The surface oxide is also fully dense, which is the opposite of the unirradiated material and more in line with irradiated and corroded material, but the fully dense nature could also be related to deformed surface observed in Figure 2(e) which may promote Cr diffusion to the surface [22]; though it is unknown how the porosity may have changed over time, but it could be speculated that the porosity closes over the long term of operation and thus the corrosion rate is reduced over time. This may correspond to the lower degree of radiolysis in the PWR environment. The radiolysis in the PWR may have been enough to increase the oxygen activity, and thus, fully densify the inner oxide and reduce the oxidation rate, but the degree of radiolysis may have been low enough to limit dissolution of the inner oxide. As a result, the overall corrosion rate goes down over time.

The combined nature of features from both irradiated and unirradiated stainless steel corrosion highlights that corrosion kinetics depend strongly on radiation flux and length of time under exposure. The mechanism for irradiation affected corrosion likely changes over time, with more dependence on radiolysis of the water, or relative lack thereof, initially and more dependence on radiation damage and RED over extended time as the oxide thickens and porosity closes. This balance in radiolysis versus RED may also account for the differences in oxide thickness and individual oxide layer thickness, particularly the nanocrystalline layer 3. BFB #4412 had a higher neutron fluence than BFB #4416 despite being in the reactor the same amount of time; the higher radiation damage or local radiolysis levels in BFB #4412 may have enhanced oxide grain nucleation before being taken over by grain growth in layer 2. Oxide grain growth may have taken place earlier in BFB #4416. This is consistent with recent research that shows susceptibility to IASCC (and thus relatedly oxidation) can vary by a factor of 2 or more depending on location of the bolts along the former plate, even when they are relatively close to each other [37]. Unfortunately, this is speculation as the ratio of layer 3 thickness between the two bolts is greater than the ratio of the neutron fluence in the two bolts, and there are potentially other effects that took place over the 40 years for the two bolts were in service. These other unknown effects could include variation in water flow near each bolt, variation in surface finish between the bolts, and potential spallation of the oxide into the water during operation.

4.2. Radiation-enhanced grain boundary attack and IASCC in commercial PWR environment

In addition to the general surface corrosion, there was substantial grain boundary attack. While general corrosion in PWR internals is less of a concern, grain boundary attack is a significant concern due to the susceptibility and effect it has on IASCC. Most studies attribute radiation induced segregation (RIS) as one of the primary reasons responsible for grain boundaries becoming more susceptible to oxidation under irradiation. This certainly appears to be the case here. The original grain boundary within the oxide is greatly depleted in Cr along with depletion in Si relative to the surrounding oxide and even the base metal, as shown in Figure 7(e). Cr is depleted, while Ni is enriched at the grain boundary. This makes the grain boundary more susceptible to oxidation due to the lack of Cr, and the Si is oxidized and quickly dissolved into the PWR water [1, 38] and thus depleted at the grain boundary within the oxide. The grain boundary ahead of the oxide in BFB #4416 in Figure 8 is actually depleted of Si, likely due to the same dissolution process into the oxide or water. The behavior of the oxidized grain boundary in the metal does not appear to be different between the harvested BFBs and prior experimental conditions, as all the key components of IASCC are present.

What appears to be new is how oxidation of the grain boundary affects oxidation at the surface of the material: the grain boundary changes the local elemental distribution within the inner oxide layers and also removes the presence of the nanocrystalline layer 3 in the vicinity of the grain boundary. The lack of Cr allows for the enhanced outward diffusion of primarily Fe to the outer oxide enabling the growth of the large Fe-rich oxide nodules on the surface that appear primarily over grain boundaries. The lack of Cr likely enables grain growth within the grain boundary oxide because it is segregated to the oxide grain boundaries; Cr appears to suppress grain growth. So not only does RIS of Ni and Si and depletion of Cr enhance susceptibility of the steel to IASCC, it also appears to enhance general surface corrosion.

An alternative possibility is that a grain boundary crack occurred as a result of IASCC and grew into the metal a few microns but stopped. Then the open crack surface went through general corrosion but on a smaller length scale closing the crack. The region along the grain boundary then

could essentially be the outermost oxide layer (layers 1 and 2 from Figure 5). So, instead of being a pathway to the outermost oxide, this effect could be two sets of oxides impinging on each other. However, this makes it difficult to explain the presence of the large, outermost Fe-rich nodules that commonly decorate the grain boundary regions.

5. Conclusions

Detailed electron microscopy characterization of the corroded surface of harvested stainless steel 316 baffle-former bolts from a commercial PWR was performed. The results showed a complex multilayered surface oxide that is in many ways different than the oxide observed in irradiated stainless steel subsequently put into simulated PWR conditions or that formed during combined ion irradiation and corrosion experiments. The results help validate and put into context previous experimental studies and will provide input for designing future research combining radiation and corrosion effects.

1. The surface oxide is $\sim 2\text{--}3\ \mu\text{m}$ thick depending on the bolt with six – not three – distinct layers that vary based on composition and grain size. The innermost layer is Cr-rich and Fe/Ni depleted, and it has a rock salt crystal structure that is radiation-induced. This sits on top of thin Ni-rich layer at the surface of the metal. The next layer is both Cr and Fe/Ni-rich with a spinel crystal structure. These two layers are formed by oxygen penetration into the metal, as interface between these and the outer oxides makes up the original metal surface. The next two layers are also likely commonly seen as the same layer, but these Cr-rich and Fe/Ni-rich oxide layers are clearly distinguished primarily by grain sizes, with layer 3 being nanocrystalline and layer 4 being above $\sim 100\ \text{nm}$ in size. Layer 3 also was shown to vary most in its thickness between the two BFBs. Lastly, the outermost oxide is a Fe-rich oxide with a large grain size of $\sim 500\ \text{nm}$. The general oxidation features corrosion behaviors in both unirradiated and ion-irradiated PWR water environments with the oxide layer phases matching that of prior irradiation studies, but the overall morphology of the surface oxide appeared to match that of unirradiated material. This could be due to the relatively low radiation flux in the BFBs compared to concurrent ion irradiation/corrosion experiments.
2. The oxidized grain boundary attack appears like the commonly found grain boundary attack in irradiation/corrosion studies with features from IASCC studies: RIS at the grain boundary, dissolution of Si within the oxide, and localized deformation. However, there is a feature that is different as the original grain boundary that is enriched in Fe/Ni and depleted in Cr appears to act as a pathway for accelerating outward diffusion of Fe/Ni to the outermost oxide layers. The grains along this path interrupt the nanocrystalline interior oxide layers as they are larger and depleted in Cr. This pathway may fuel the growth of the outermost oxide nodules that decorate grain boundaries and may further enhance general oxidation and grain boundary attack.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by the US Department of Energy, Office of Nuclear Energy, Light Water Reactor Sustainability Program Materials Research Pathway under contract DE-AC05-00OR22725 with UT-Battelle LLC via Oak Ridge National Laboratory. The authors would like to thank Patricia Tedder and Travis Dixon at the Low Activation Materials Development and Analysis (LAMDA) Laboratory at ORNL, where SEM and TEM/STEM evaluation was conducted. Microscopy in LAMDA is a cost recovery center available for users through the US-DOE Nuclear Science User Facilities. The initial handling and cutting of the bolts were done in the hot cells in the Irradiated Materials Examination and Testing (IMET) facility at ORNL. The Thermo Fisher Talos 200X STEM in LAMDA was purchased and funded through the US-DOE Office of Nuclear Energy Advanced Fuel Campaign and Nuclear Science User Facilities. The authors would like to thank the late Mike Burke who was involved in the bolt harvesting and characterization planning and led the sample preparation, machining, and shipping when he was working at Westinghouse Electric Company.

Data Availability

Data is available upon request.

References

1. Was, G.S., et al., *How irradiation promotes intergranular stress corrosion crack initiation*. Progress in Materials Science, 2024. **143**: p. 101255.
2. Shoji, T., S.-i. Suzuki, and K.S. Raja, *Current status and future of IASCC research*. Journal of Nuclear Materials, 1998. **258-263**: p. 241-251.
3. Ickes, M.R., et al., *Irradiation-assisted stress corrosion cracking of Type 347 and Type 316 steels irradiated in commercial pressurized water reactors*. Journal of Nuclear Materials, 2020. **536**: p. 152182.
4. Wang, S., et al., *Clarifying the mitigation effect of proton irradiation on the intergranular oxidation of 316L stainless steel in high temperature water*. Acta Materialia, 2022. **241**: p. 118408.
5. Du, D., K. Sun, and G.S. Was, *IASCC of neutron irradiated 316 stainless steel to 125 dpa*. Materials Characterization, 2021. **173**: p. 110897.
6. Wang, S., et al., *The time dependence of proton irradiation effect on the intergranular oxidation of 316 L stainless steel in high-temperature hydrogenated water*. Acta Materialia, 2023. **260**: p. 119340.
7. Wang, S., et al., *Clarifying the irradiation effect on the general oxidation of 316L stainless steel in high temperature hydrogenated water after 1000h immersion*. Acta Materialia, 2023. **255**: p. 119100.
8. Boisson, M., et al., *Role of irradiation and irradiation defects on the oxidation first stages of a 316L austenitic stainless steel*. Corrosion Science, 2019. **161**: p. 108194.
9. Deng, P., et al., *Effect of irradiation on corrosion of 304 nuclear grade stainless steel in simulated PWR primary water*. Corrosion Science, 2017. **127**: p. 91-100.
10. Perrin, S., et al., *Influence of Irradiation on the Oxide Film Formed on 316 L Stainless Steel in PWR Primary Water*. Oxidation of Metals, 2013. **80**(5): p. 623-633.
11. Raiman, S.S. and G.S. Was, *Accelerated corrosion and oxide dissolution in 316L stainless steel irradiated in situ in high temperature water*. Journal of Nuclear Materials, 2017. **493**: p. 207-218.
12. Raiman, S.S., D.M. Bartels, and G.S. Was, *Radiolysis driven changes to oxide stability during irradiation-corrosion of 316L stainless steel in high temperature water*. Journal of Nuclear Materials, 2017. **493**: p. 40-52.

13. Liu, H., G.-H. Lei, and H.-F. Huang, *Review on synergistic damage effect of irradiation and corrosion on reactor structural alloys*. Nuclear Science and Techniques, 2024. **35**(3): p. 57.
14. Wang, P., et al., *Effect of radiation damage and water radiolysis on corrosion of FeCrAl alloys in hydrogenated water*. Journal of Nuclear Materials, 2020. **533**: p. 152108.
15. Scott, P., *A review of irradiation assisted stress corrosion cracking*. Journal of Nuclear Materials, 1994. **211**(2): p. 101-122.
16. Lach, T.G. and X. Chen, *Post Irradiation Examination of Pressurized Water Reactor Stainless Steel Internal Components*, in *ASME 2023 Pressure Vessels & Piping Conference, PVP2023*. p. 107347.
17. Ishiyama, Y., R.B. Rogge, and M. Obata, *Radiation-induced stress relaxation in high temperature water of type 316L stainless steel evaluated by neutron diffraction*. Journal of Nuclear Materials, 2011. **408**(2): p. 153-160.
18. Mogren, T., *Response to Request for Additional Information RE: The 2011 Steam Generator Tube Inspections - R.E. Ginna Nuclear Power Plant*, U.S. NRC, Editor. 2012, CENG, LLC: www.nrc.gov/docs/ML1205/ML12053A147.
19. Najafkhani, F., et al., *Recent advances in the kinetics of normal/abnormal grain growth: a review*. Archives of Civil and Mechanical Engineering, 2021. **21**(1): p. 29.
20. Chang, L., et al., *Effect of machining on stress corrosion crack initiation in warm-forged type 304L stainless steel in high temperature water*. Acta Materialia, 2019. **165**: p. 203-214.
21. Qian, H., et al., *The effect of surface grinding on the stress corrosion cracking initiation of 316LN stainless steel in 600 °C supercritical CO₂*. Corrosion Science, 2024. **234**: p. 112147.
22. Das, A., S. Roychowdhury, and V. Kain, *Effect of surface state of austenitic type 304L SS on oxide characteristics formed in high temperature high pressure water*. Journal of Nuclear Materials, 2022. **564**: p. 153672.
23. Lach, T.G., M.N. Gussev, and X. Chen, *Complexity of segregation behavior at localized deformation sites formed while in service in a 316 stainless steel baffle-former bolt*. Scripta Materialia, 2025. **255**: p. 116371.
24. Terachi, T., et al., *Corrosion Behavior of Stainless Steels in Simulated PWR Primary Water—Effect of Chromium Content in Alloys and Dissolved Hydrogen—*. Journal of Nuclear Science and Technology, 2008. **45**(10): p. 975-984.
25. Terachi, T., K. Fujii, and K. Arioka, *Microstructural Characterization of SCC Crack Tip and Oxide Film for SUS 316 Stainless Steel in Simulated PWR Primary Water at 320°C*. Journal of Nuclear Science and Technology, 2005. **42**(2): p. 225-232.
26. Hemmi, Y., Y. Uruma, and N. Ichikawa, *General Corrosion of Materials under Simulated BWR Primary Water Conditions*. Journal of Nuclear Science and Technology, 1994. **31**(5): p. 443-455.
27. Chen, K., et al., *A high-resolution study of the different surface state effects on the corrosion behaviors of a ferritic steel and an austenitic steel in supercritical water*. Corrosion Science, 2022. **209**: p. 110757.
28. Hanbury, R.D. and G.S. Was, *Oxide growth and dissolution on 316L stainless steel during irradiation in high temperature water*. Corrosion Science, 2019. **157**: p. 305-311.
29. Barr, C.M., et al., *Anisotropic radiation-induced segregation in 316L austenitic stainless steel with grain boundary character*. Acta Materialia, 2014. **67**: p. 145-155.
30. Bosch, R.W., et al., *Crack initiation testing of thimble tube material under PWR conditions to determine a stress threshold for IASCC*. Journal of Nuclear Materials, 2015. **461**: p. 112-121.
31. Gussev, M.N., K.G. Field, and J.T. Busby, *Deformation localization and dislocation channel dynamics in neutron-irradiated austenitic stainless steels*. Journal of Nuclear Materials, 2015. **460**: p. 139-152.
32. Hanbury, R.D. and G.S. Was, *Corrosion of 316L stainless steel in high temperature water and steam: Mechanisms of corrosion*. Journal of Nuclear Materials, 2024. **599**: p. 155180.
33. Hanbury, R.D. and G.S. Was, *Corrosion of 316L stainless steel in high temperature water and steam: effects of simultaneous proton irradiation*. Journal of Nuclear Materials, 2024. **600**: p. 155274.

34. Lee, H.B., F.B. Prinz, and W. Cai, *Atomistic simulations of grain boundary segregation in nanocrystalline yttria-stabilized zirconia and gadolinia-doped ceria solid oxide electrolytes*. Acta Materialia, 2013. **61**(10): p. 3872-3887.
35. Wei, S., et al., *Radiation induced segregation in titanium diboride*. Acta Materialia, 2024. **267**: p. 119739.
36. Hosemann, P., et al., *Characterization of oxide layers grown on D9 austenitic stainless steel in lead bismuth eutectic*. Journal of Nuclear Materials, 2008. **375**(3): p. 323-330.
37. Jyung, J.-M., et al., *Irradiation-induced age-related degradation mechanisms effect on integrity of baffle former bolts*. International Journal of Pressure Vessels and Piping, 2024. **208**: p. 105116.
38. Kuang, W., et al., *A high-resolution characterization of irradiation-assisted stress corrosion cracking of proton-irradiated 316L stainless steel in simulated pressurized water reactor primary water*. Corrosion Science, 2022. **199**: p. 110187.